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Team Details:

Team name :CODE CRACKERS

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Track Details:

Track name : BIOTECH TRACK

Problem statement : AI COPILOT FOR SCALABLE CELL-CULTURE BIOPROCESS DESIGN

Problem Statement

- High media and growth-factor cost. Oxgyen transfer, mixing and shear do not scale linearly with vessel volume. Glucose overfeeding can drive lactate, amino -acid metabolism can increase ammonia.

- Cellular-agricultural terms, cultured food developers process-development labs and bioreactor operators. The same design must work from bench experiments to pilot/production scale.

- A process that works at 2L can fail at 1,000 L because KLA, gas dispersion ,power input and local concentration gradients change. Cost and product quality must remain controlled together.

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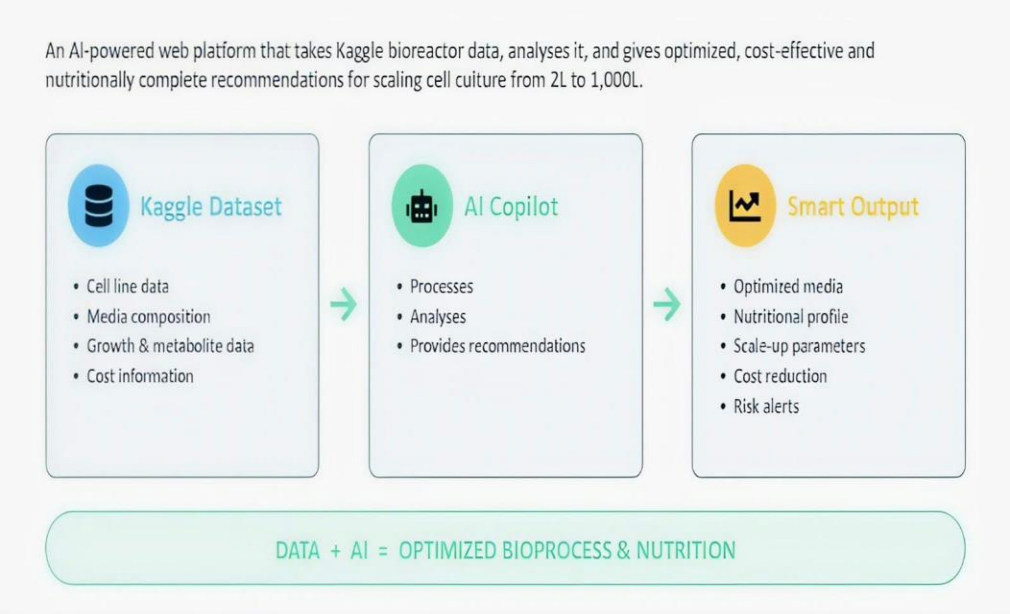
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Proposed Solution

- Import cell-culture/bioreactor databases from kaggle. Use parameters such as glucose, glutamine, lactate, ammonia, VCD, viablity, pH and DO.
- Validate and clean the incoming data.
- Compare the user's current nutrient levels with historical/reference data.
- Detect nutrient deficiency, excess feeding and abnormal metabolite trends.
- Predict how nutrient changes may affect cell growth and viability.



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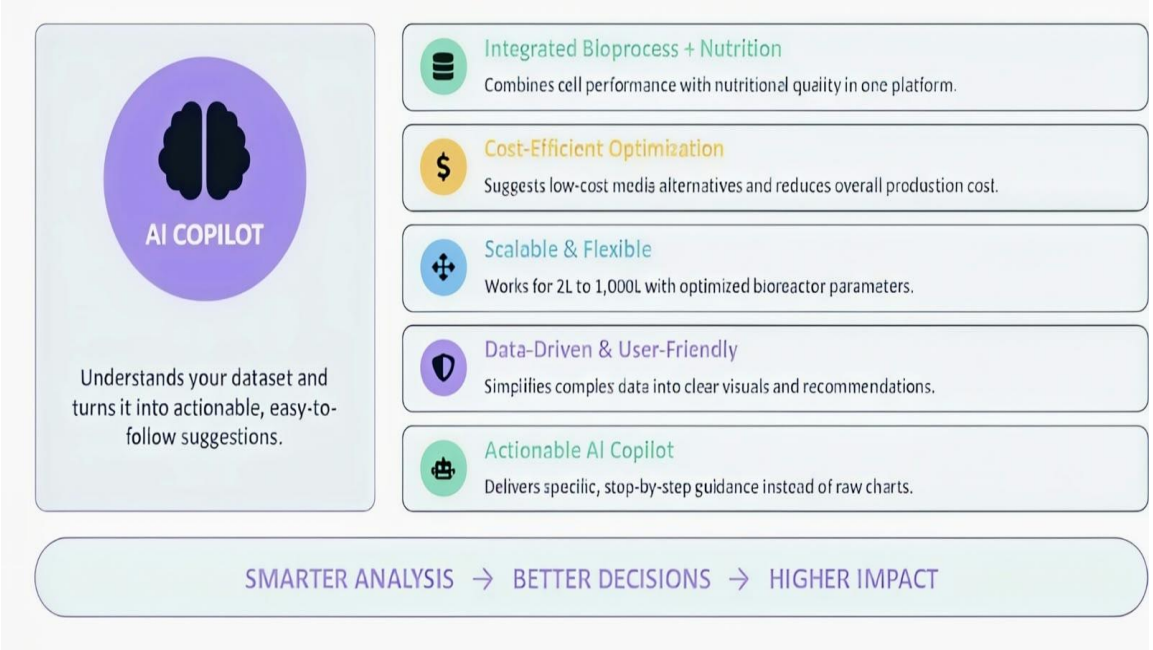
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Innovation / Uniqueness

1. AI-powered nutrient optimization
2. Kaggle data + AI
3. Cost + growth optimization together
4. Predictive metabolic analysis
5. 2 L → 1,000 L scale-up support
6. Explainable AI Copilot



Target Users

- Bioprocess R&D teams can use the Copilot to optimize cell-culture media and feeding strategies.
- Cellular agriculture companies can use it to reduce production costs while maintaining cell growth and viability.
- Process engineers can use it to evaluate scale-up conditions from 2 L to 1,000 L.
- Food scientists can use it to evaluate nutritional requirements and formulations.
- Manufacturing and QA teams can use it to monitor process performance and deviations.

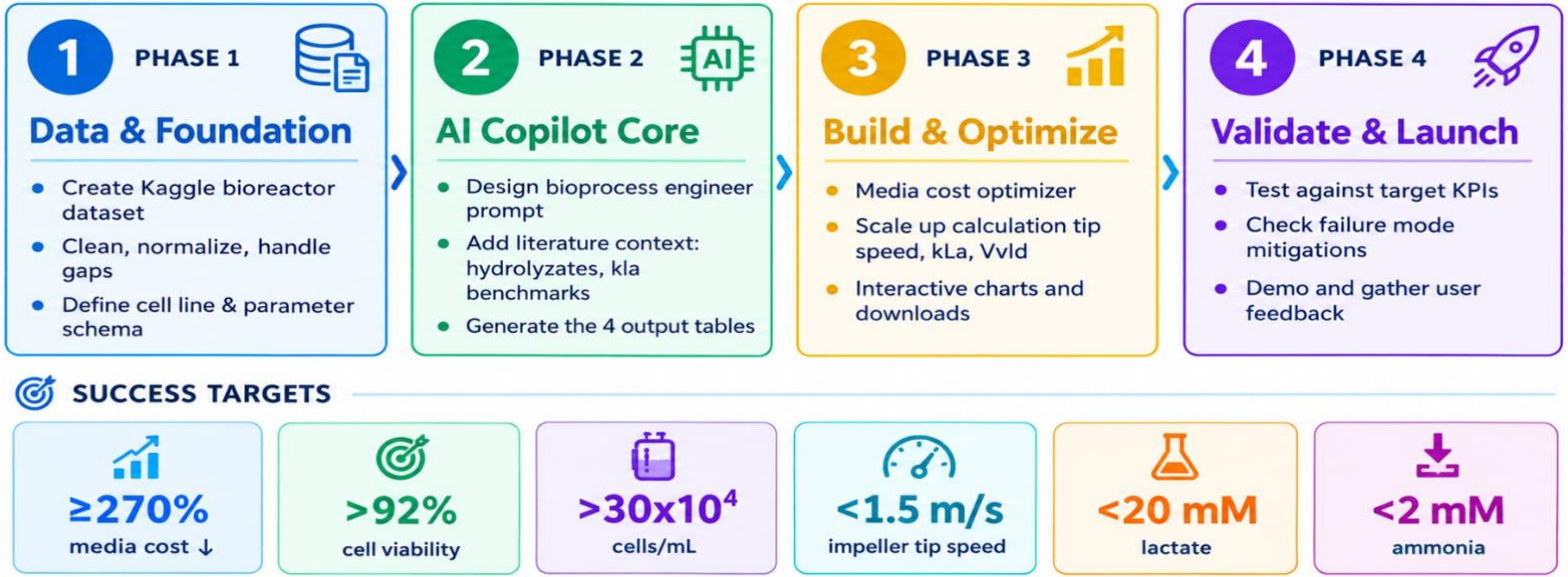
Key Features

- Media Cost Optimizer – Identifies lower-cost nutrient and feed alternatives.
- Nutritional Analysis – Checks essential nutrients required for cell growth.
- AI Metabolic Prediction – Predicts growth and metabolite trends.
- Scale-Up Calculator – Supports scale-up from 2 L to 1,000 L.
- Smart Feed Scheduler – Recommends nutrient feeding rates and timing.
- Failure Detection – Identifies risks such as nutrient deficiency, lactate accumulation, oxygen limitation, and shear stress.

Technology / Tech Stack

1. Languages : Python, JavaScript
2. Frameworks : React.js, FastAPI
3. AI/ML Models : Gradient Boosting, Regression, Bayesian Optimization
4. APIs : REST API, FastAPI
5. Cloud/Platform : Google Colab / Kaggle, Vercel
6. Software/Tools : PostgreSQL/SQLite, Pandas, NumPy, Plotly, GitHub

Implementation Plan



Expected Impact

- Reduce media cost through nutrient and feed optimization.
- Improve cell viability and growth through better nutrient control.
- Reduce lactate and ammonia accumulation by optimizing feeding.
- Support 2 L → 1,000 L scale-up with data-driven process guidance.
- Improve nutritional quality through nutrient-profile monitoring.

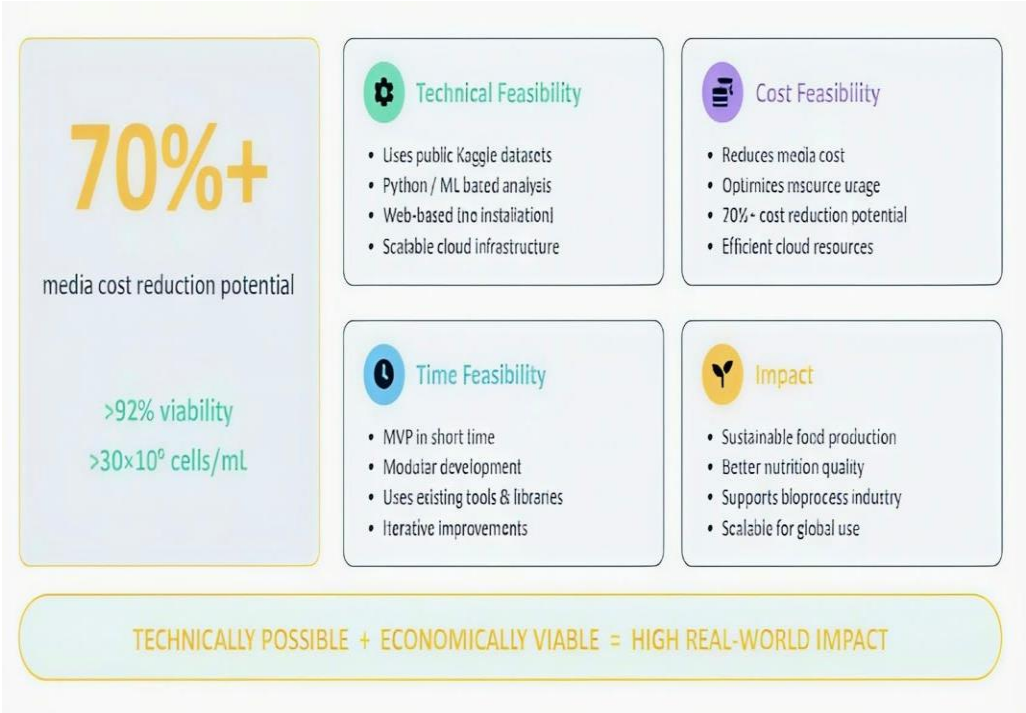
Feasibility

1.Data Feasibility: Use Kaggle datasets and literature data to build the AI model.

2.Technical Feasibility: Implement the solution using Python, AI/ML, FastAPI, React and a database.

3.Operational Feasibility: Provide nutrient and process recommendations through a web dashboard.

4.Validation: Test AI recommendations against process constraints before human approval.



Prototype / References (optional)

Figma

GitHub

Demo

Architecture diagram

Research/reference links

One-Line Pitch

An AI-powered Bioprocess Copilot that uses cell-culture data to optimize nutrients, reduce media costs, improve cell growth, and guide scalable production from 2 L to 1,000 L.